AP Statistics — 1023 Quiz 2 · FRQ

Section II: Free Response · Questions 1–3

Directions: Show all your work. Indicate clearly the methods you use, because you will be scored on the correctness of your methods as well as on the accuracy and completeness of your results and explanations. Write all of your answers on the Answer Sheet; work written in this booklet will not be scored. Each question is scored from 0 to 4.

Question 1 (4 points)

The table below shows the political party registration by gender of all 500 registered voters in Franklin Township.

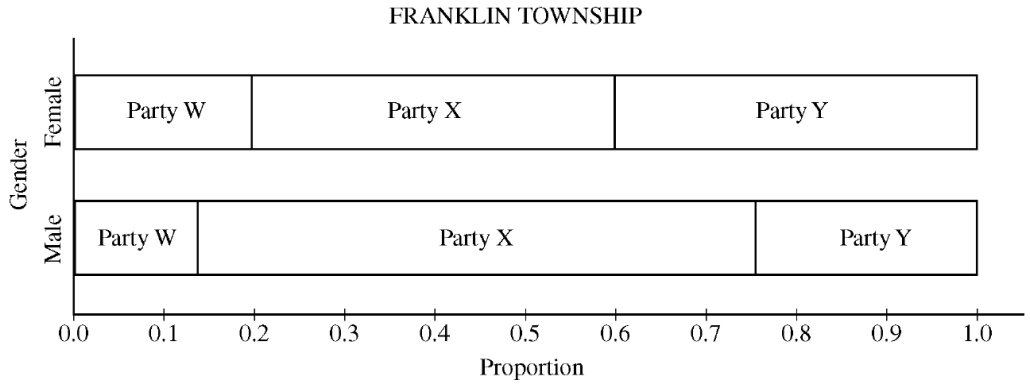
PARTY REGISTRATION–FRANKLIN TOWNSHIP

	Party W	Party X	Party Y	Total
Female	60	120	120	300
Male	28	124	48	200
Total	88	244	168	500

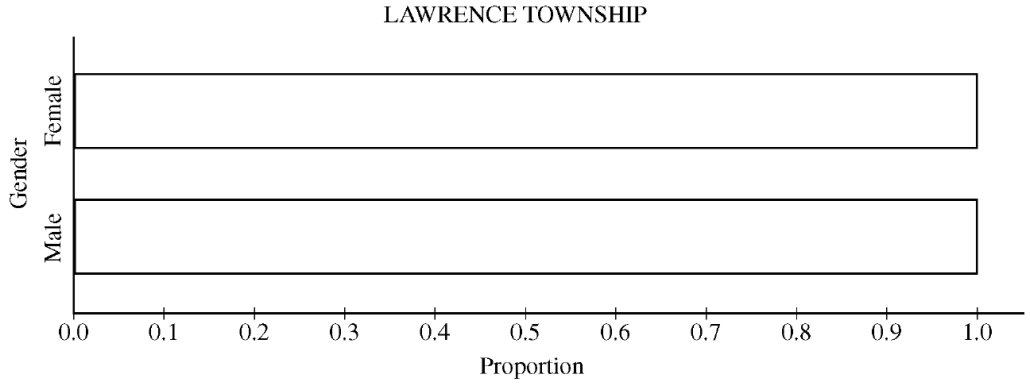
(a) Given that a randomly selected registered voter is a male, what is the probability that he is registered for Party Y?

(b) Among the registered voters of Franklin Township, are the events "is a male" and "is registered for Party Y" independent? Justify your answer based on probabilities calculated from the table above.

(c) One way to display the data in the table is to use a segmented bar graph. The following segmented bar graph, constructed from the data in the party registration–Franklin Township table, shows party-registration distributions for males and females in Franklin Township.



In Lawrence Township, the proportions of all registered voters for Parties W, X, and Y are the same as for Franklin Township, and party registration is independent of gender. Complete the graph below to show the distributions of party registration by gender in Lawrence Township.



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Question 2 (4 points)

Nine sales representatives, 6 men and 3 women, at a small company wanted to attend a national convention. There were only enough travel funds to send 3 people. The manager selected 3 people to attend and stated that the people were selected at random. The 3 people selected were women. There were concerns that no men were selected to attend the convention.

(a) Calculate the probability that randomly selecting 3 people from a group of 6 men and 3 women will result in selecting 3 women.

(b) Based on your answer to part (a), is there reason to doubt the manager's claim that the 3 people were selected at random? Explain.

(c) An alternative to calculating the exact probability is to conduct a simulation to estimate the probability. A proposed simulation process is described below.

> Each trial in the simulation consists of rolling three fair, six-sided dice, one die for each of the convention attendees. For each die, rolling a 1, 2, 3, or 4 represents selecting a man; rolling a 5 or 6 represents selecting a woman. After 1,000 trials, the number of times the dice indicate selecting 3 women is recorded.

Does the proposed process correctly simulate the random selection of 3 women from a group of 9 people consisting of 6 men and 3 women? Explain why or why not.

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Question 3 (4 points)

The travel time for the 7:40 a.m. commuter train from Oakdale to Central Station is approximately normally distributed with mean 32 minutes and standard deviation 2.5 minutes. A trip is considered late if the travel time is more than 36 minutes.

(a) What is the probability that a randomly selected trip is late? Show your work.

(b) Mia takes this train on 5 randomly selected days. Assuming that travel times on different days are independent, find the probability that at least one of Mia's 5 trips is late.

(c) For one randomly selected trip, let A be the event "the trip is late" and let B be the event "the travel time is less than 30 minutes."

  (i) Are events A and B mutually exclusive? Explain.

  (ii) Are events A and B independent? Justify your answer using probabilities.

(d) Suppose the probability that a trip is late is 0.05. Describe how to use a table of random digits to carry out one trial of a simulation that estimates the probability that at least one of 5 trips is late.